Analyzing images on social media for (not) using mask

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BTP Track: Research

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Student's Declaration

I hereby declare that the work presented in the report entitled **Analyzing images on social media for (not) using mask** submitted by us for the partial fulfillment of the requirements for the degree of *Bachelor of Technology* in *Computer Science & Engineering* at Indraprastha Institute of Information Technology, Delhi, is an authentic record of our work carried out under guidance of **Dr. Ponnurangam Kumaraguru and Dr. Tavpritesh Sethi**. Due acknowledgements have been given in the report to all material used. This work has not been submitted anywhere else for the reward of any other degree.

Asmit/Paras **Place & Date: New Delhi, 7th May 2021**

Certificate

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Dr. Ponnurangam Kumaraguru **Place & Date: New Delhi, 7th May 2021**

Dr. Tavpritesh Sethi

Abstract

The adoption of non-pharmaceutical interventions and their surveillance is critical for detecting and stopping possible transmission routes of COVID-19. A study of the effects of these interventions in terms of adoption can help shape public health decisions. Social media analytics can offer crucial public health insights. In this thesis we examined mask use and mask fit by analyzing 2.04 million posts from the 6 United States cities, 0.19 million posts during a large scale public event, Black Lives Matter (BLM) and compared how Democrats and Republicans differ in their behaviour towards NPIs, in the light of 2020 presidential elections using 3.9 million posts.

The study finds a significant drop in the group posting when the stay-at-home laws were applied and a significant increase in mask wearing for two of the three cities when the mask mandates were applied. Although a general positive trend towards mask-wearing and social distancing is observed, a high percentage of posts did not adhere to the guidelines. BLM related posts were found to capture the lack of seriousness to safety measures through a high percentage of group pictures and low mask fit scores. Posts from Democratic states were found to have significantly higher percentages of detected masks, and significantly lower percentages of group photos, showing higher adoption of mask-wearing and social distancing measures than Republican states. Thus, the methodology used provides a directional indication of how government policies can be indirectly monitored through social media.

Keywords: *Covid-19, Mask Detection, Deep Learning Classification, Segmentation, Social Media Analysis*

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Work Distribution

RQ 1,2,3,4 from were covered in the first semester of the Btech project, While RQ 5 and 6 were covered in the second semester. Chapter 5 was additionally developed during the second semester.

Contents

1 Introduction **1**

1.1 Overview 1

1.2 Related Works 2

1.3 Research Questions 3

2 Datasets **4**

2.1 Mask-Unmask Classifier Dataset 4

2.2 Fit Score Dataset 5

2.3 USA-Cities Instagram dataset 5

2.4 Democrats vs Republican Dataset 5

3 Proposed Frameworks **8**

3.1 Mask-Unmask classification framework 9

3.2 Fit score analysis framework 11

4 Analysis **13**

4.1 City-wise Analysis 13

4.2 Impact of Government Policies 16

4.2.1 Stay At Home Laws 17

4.2.2 Mask Wearing mandates 17

4.3 Protests and the Pandemic 19

4.4 Democrats vs Republican Analysis 21

5 Analysis tool **24**

5.1 Pages 24

5.1.1 Explore Page 24

5.1.2 Request Page 24

5.1.3 Report Page 26

5.2 Data Pipeline 26

6 Conclusion **29**

6.1 Limitation 30

6.2 Future Works 31

6.3 FAIR 31

Chapter 1

Introduction

1.1 Overview

The outbreak of COVID-19 has the world in its grips. The World Health Organisation (WHO) declared it as a global pandemic on March 11, 2020 [48], and with exponentially rising cases, there are currently more than 32 million cases and 500,000 deaths(April 2021) [9]. Following WHO health advisories, most countries have declared national emergencies, closed borders, and restricted public movement [47, 49]. Masks have been found to reduce potential exposure risk from an infected person, proving to be a successful measure to suppress transmission and save lives [20, 23, 34, 52]. Many studies have recognized the importance of community-wide wearing masks for controlling the pandemic [19, 63]. Social distancing measures were also applied to prevent sick individuals from coming in contact with healthy individuals. These social distancing measures and mask-wearing have proven successful in many countries like China [29,32]. Governments worldwide have used social distancing and mask-wearing as a primary non-pharmaceutical measure against the virus.

With over 32 million cases and 500,000 deaths(as of April, 2021), the USA is one of the largest hit countries by the virus. In the USA, many state governments had applied several stay-at-home and mask-wearing measures as early non-pharmaceutical interventions. In the lead-up to widespread vaccine deployment, the adoption of non-pharmaceutical interventions and their surveillance is critical for detecting and stopping possible transmission routes. Quantifying the effectiveness of such measures is a challenging task, which currently relies on on-ground surveys [45] or self-reported numbers [52]. During the pandemic, the country also observed the Black Lives Matter protests. The killing of George Floyd on May 25 2020, sparking a series of

protests [60] and agitation across the country. Protests are designed to stimulate public action for social justice. Such protests involve the physical gathering of people, making it difficult to adhere to social distancing. These protests provide an opportunity to observe how people react to public health-related preventive measures during such gatherings, but collecting the necessary data from the ground is very challenging given the circumstances. Such protests have also gained high popularity and attraction through online social media [25, 46].

In the lead-up to the 2020 presidential elections, mask-wearing and social distancing had become a partisan issue. According to Pew Research, people who lean Democratic are more likely than Republicans to say that they wore a mask, with conservative Republicans, being amongst the least likely to say they wore a mask, while liberal Democrats being amongst the most likely [2]. A significant difference was found between the percentage of people saying they wore a mask vs the percentage of people who said they saw others wearing a mask in the USA. Hence, quantifying the number of people actually wearing face-covering or following social distancing is a challenging task, which currently relies on on-ground surveys, or self-reported numbers [1]. Through this research, we move to social media for answers.

With its ease of access and global outreach, social media has a disproportionate influence on disseminating information during a pandemic [28]. In recent times of the pandemic, social media has become a popular platform for people to express their thoughts, opinions, and broadcast activities. The general public and authorities have been using hashtags like #CoronaOutbreak, #COVID19, and #mask to disseminate important information and health advisories, which provides us with an opportunity to analyze behaviours and the impact of such advisories worldwide [12].

1.2 Related Works

Social media has been extensively explored and analyzed for patterns that have emerged during the COVID-19 pandemic [13,22,35,37]. Signorini et al. [58] examined Twitter based-information to track the swiftly-evolving public sentiment regarding Swine Flu in 2011 and correlate the H1N1 virus subtype-related activity to track reported disease levels in the US accurately. Realizing the potential of social media in understanding such events, in this study, we utilize the social media images from Instagram, a popular image-sharing social media platform. Computer vision-based classifiers are necessary to check if a person is wearing a mask from its image.

There exist two datasets previously published for mask classification tasks, MAFA(MAsked FAces) [31] and RMFD(Real world Masked Face Dataset) [38]. MAFA dataset contains 35,805 masked images. Since the MAFA Dataset was curated and released in 2017, it couldn't capture different varieties and types of masks that have been in use during the pandemic period. The MAFA dataset is biased towards one kind of mask; it majorly consists of medical staff wearing disposable medical-grade masks. RMFD contains 7,959 masked images containing a variety of masks used during the COVID-19 period. However, manual qualitative evaluation of the images revealed that the images were not suitable for analyzing high-quality social media images since the majority of the images were of less than 50x50 resolution after cropping their face region. In addition to mask detection, analyzing the fit of the mask is a highly useful application. There is no previous work that is trying to analyze mask fit using semantic segmentation and a corresponding annotated dataset for the same to the best of our knowledge.

1.3 Research Questions

The following research questions were explored when conducting research for this project:

1. How many people are wearing masks in images on social media, and what is their fit?
2. Is there any correlation between the COVID-19 cases and the number of masks worn when looked through online social media?
3. How successful are the stay-at-home laws and mask-wearing mandates put by governments as seen online?
4. Was there any change in mask wearing behaviour of people while they were out in protests?
5. Do Democrats and Republicans have a significant difference in their adherence to mask-wearing as seen through their social media images?
6. Do Democrats and Republicans any significant difference in their adherence to social distancing as seen through their social media images?

Chapter 2

Datasets

The study uses four different dataset collections for different purposes.

2.1 Mask-Unmask Classifier Dataset

This dataset is used for training a model that classifies whether the person in the image is wearing a mask or not. So we needed images in which people were wearing masks (masked images) and images in which people were not wearing masks (unmasked images) so that our model can learn to distinguish between the two categories. We collected around 30,000 images of people wearing masks from Google Search Images using tags "people wearing masks", "children wearing masks". Images from Instagram were also collected with the Tag explore feature, using these three tags: "mask", "masked", "covidmask". Although the tagging algorithms used by Google is expected to capture most of the images, some images may have been potentially missed. There is further scope for expanding our set of tags chosen to capture the entire population of images in which people are wearing masks, and this is a limitation of our current approach. However, this will need more research as capturing other scenarios may also lead to noisier sets. Post data collection, images with a width and height of at least 50 pixels were kept to ensure decent image quality. Then Face Detector was used on these images to extract faces. The images of extracted faces were distributed between 5 annotators, and annotators were asked to classify the face as either "masked" or "unmasked". After the annotations, images of 9,055 masked faces were obtained. This dataset of 9,055 masked images (VAriety MAsks - Classification (VAMA-C)) is publicly available for use in further research [3]. For un-masked face images, we made a random sample(without replacement) of size 9,055 from the VGGFace2 Dataset [18], which is a

large-scale face recognition dataset. Sample from VGGFace2, along with VAMA-C, was used to train the Unmask-Mask Classifier, whose details are given in the Proposed Frameworks section.

2.2 Fit Score Dataset

Out of 9,055 masked faces that we obtained from the previous dataset, we selected 504 images with different poses and a wide variety of mask designs. Then we annotated these images using Label Studio [2] for getting pixel-level annotations of mask region on the face. This dataset was then used to train a semantic-segmentation model, whose details are in section 3.2. This dataset (VAriety MAsks - Segmentation (VAMA-S)) is publicly available for use in further research [3].

2.3 USA-Cities Instagram dataset

For the analysis phase, we collected the location-tagged public posts from Instagram between February 1, 2020, and May 31, 2020, for six cities New York City, Seattle, Dallas, New Orleans, Minneapolis, and Boston. The first covid case was reported in January 2020 [33], and till July 2020 US is still in the first wave of the coronavirus pandemic [24]. Hence, the chosen time frame captures the beginning and growth of the US's COVID-19 pandemic. This collection was done for six major US cities. These six cities were selected to represent different geographical sections of the country. We collected a total of 2.04 million public posts from these six cities. These posts were collected via Instagram's explore location feature. Instagram's GraphQL API was employed for the data collection. The tools used have been published as a python PyPI package [4]. The states and the summary statistics can be found in the (Table 2.1). We also collected data for New York City and Minneapolis from May 25, 2020, to July 15, 2020. These two cities observed major protests [11]. We curate a list of trending tags and keywords ('blm', 'blacklivesmatter', ' georgefloyd', 'justiceforgeorgefloyd', 'policebrutality', and 'protest') during this period. We refer to the posts whose captions included these tags as BLM posts and the rest as Non-BLM posts.

2.4 Democrats vs Republican Dataset

We collected public Instagram posts for the 4 most democratic states (Vermont, Maryland, California, and Massachusetts), and 4 most republican states(Wyoming, Oklahoma, West-Virginia

City	Collected posts
New York City	245,677
Dallas	540,500
Seattle	437,040
Minneapolis	220,999
New Orleans	315,082
Boston	283,757
Total Images	2,043,055

Table 2.1: Number of posts collected for each city from Instagram. Most posts were collected for Dallas, while the least for Minneapolis

and North-Dakota) by vote percentages in the 2020 presidential elections between 1st March 2020 and 1st December 2020, in order to capture the buildup to the elections. The data was collected through crawling Instagram's explore location page. For a given state we collected data for the state's location along with locations for some of the most populous cities in the state. The figure presents the location wise data collected. We collected a total of 3,917,955 posts of which 78.3% (3,067,740) of the posts were collected from Democratic states, while 21.7%(850,215) of the posts were collected from the Republican states. This disparity in the number of posts is in line with the low number of Instagram users from the top 4 Republican states as compared to the top 4 Democratic states. Since our analysis is based on ratio and percentages, this difference in number was not a major issue. The data distribution among the cities is represented in Figure 2.1

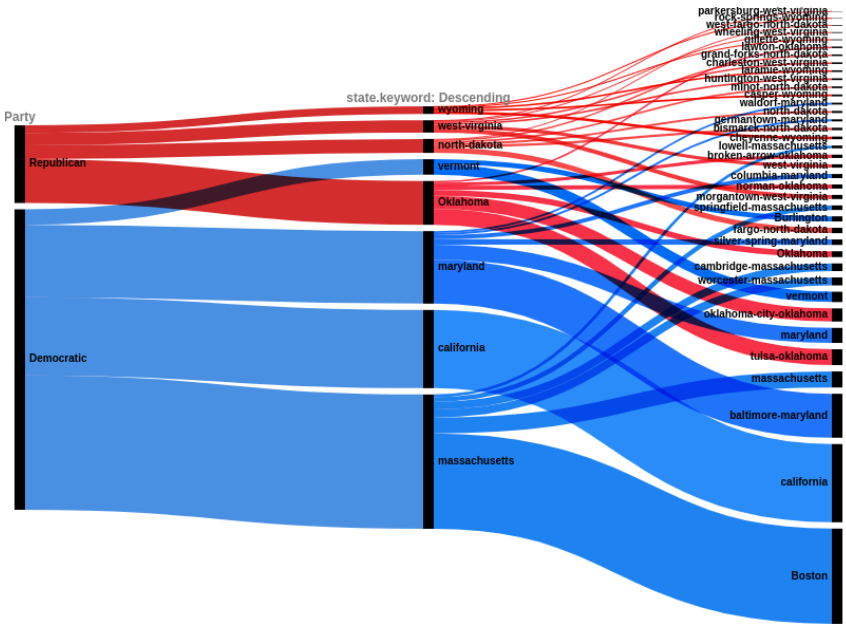

Figure 2.1: Alluvial Diagram showing the collected dataset. The colour of the bars represents the percentage vote given to the respective political party, red being Republican and blue being Democratic. The size of the bars represents the number of collected posts from the specific location. The 3rd column is sorted by the number of posts collected from the location.

Chapter 3

Proposed Frameworks

This article proposes a Mask-Unmask classification framework (for classifying masked and unmasked images) and a fit score analysis framework (for evaluating whether the masks are being worn effectively or not from the given image). The Mask-Unmask classification model is used to analyze the dataset "USA-Cities Instagram dataset" and "Democrats vs Republican Dataset", while the Fit-score analysis framework is just used for BLM posts, in order to capture the mask wearing patterns during a huge social gathering.

Images obtained from sources mentioned in the previous section consisted of various individuals. Thereby, to detect face masks in these images, both the framework's first task was face detection. This was done using the pre-trained model Retinaface [27], which is one of the top-performing models on Face Detection on the WIDER Face(Hard) dataset [64]. Next, facial landmarks were obtained using the Dlib's implementation [42] of finding facial landmarks(Figure 1) proposed by Kazemi et al. [39]. The landmark nos. 5-13, 31-36, 49-68 were used to filter the face's jaw region. This jaw region obtained was then used as input to the classification model for the mask-unmask classification framework. The landmark nos. 32-36, 49-68 was used to filter the

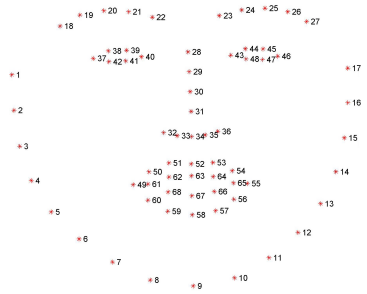

Figure 3.1: Facial Landmarks detected on a face using DLib

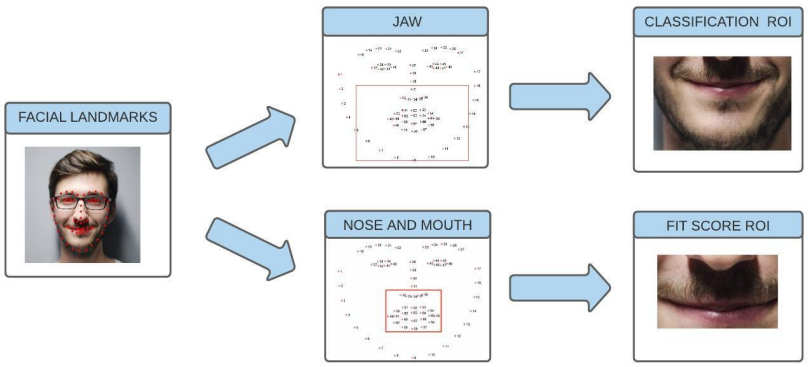

Figure 3.2: Face Mask Detection Pipeline. Given the input image, it detects faces, then extracts jaws from each of the faces, then classifies the jaw as being masked and unmasked.

nose-mouth region from the face. This nose-mouth region obtained was then used for calculating the fit-score. An example of an output of a face's facial landmarks is shown in (Figure 3.1).

3.1 Mask-Unmask classification framework

After obtaining the image of the jaw, the images were classified depending on whether the jaw region contained a mask over it or not (Figure 3.3). The following architectures were experimented with while training a classification model - MobileNet V2, Nas Net, EffecientNet B0, EffecientNet B1, EffecientNet B2, DenseNet121. These models were selected since they have significantly fewer parameters than most other architectures (Table 3.1). The input image size for all the models was 224 x 224. Transfer learning was used, and weight initialization of all models was done using ImageNet. All the models were truncated at the last fully connected layer. The following layers were added: (i) Average Pooling with 5 x 5 Pool size, (ii) Flatten Layer, (iii) Dense Layer with 128 hidden units and reLU activation (iv) Dropout layer of 0.5, and (v) Dense Layer of 2 hidden units and Softmax Activation. Adam optimizer with an initial learning rate of 1e-4 was used, and each model was trained for 30 epochs with a batch size of 64, with binary cross-entropy as the loss function. For training the classifier for detecting mask and unmask from the input jaw regions, the total data from Section. 2.1 consisted of 9,055 masked and 9,055 unmasked samples, which were split into 80:20 ratio of train and validation set. 5-Fold Cross-validation was used to evaluate the trained models' performance, and the results are shown in (Table 3.1).

The evaluation indicates that EfficientNet B0 emerges as the best performing model with an overall accuracy of 0.98 $\pm$ 0.01. EfficientNet [14] is a convolutional neural network architecture

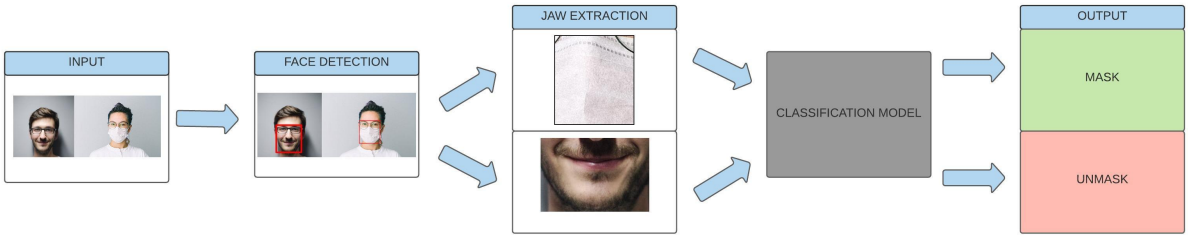

Figure 3.3: Face Mask Detection Pipeline

Model	No. of Parameters	Recall	Accuracy
MobileNet V2	3,538,984	0.90 ± 0.01	0.94 ± 0.01
Nas Net Mobile	5,326,716	0.86 ± 0.01	0.91 ± 0.01
EfficientNet B0	5,330,571	0.97 ± 0.01	0.98 ± 0.00
EfficientNet B1	7,856,239	0.97 ± 0.01	0.98 ± 0.00
DenseNet121	8,062,504	0.90 ± 0.01	0.94 ± 0.01
EfficientNet B2	9,177,569	0.97 ± 0.01	0.98 ± 0.00

Table 3.1: Face Mask Detection Model Results. EffecientNet B0 is used as the model for face mask detection due to its superior performance and smaller model size. The precision for each model was found to be 1.00 ± 0.01

and scaling method that uniformly scales all depth/width/resolution dimensions. Efficient B0 has 5.3 Million parameters with 18 layers. Its architecture consists of an initial 3x3 convolution layer, followed by a series of MBconv layers with different kernel sizes and no. of channels. The series of MBconv layers are followed by a convolution layer, a pooling layer, and a fully connected layer. It uses linear activation in the last layer in each block to prevent loss of information from ReLU. Compared to the conventional CNN models, the main building block for EfficientNet is MBConv, an inverted bottleneck conv, known initially as MobileNetV2. Before the EfficientNets came along, the most common way to scale up ConvNets was either by one of three dimensions - depth (number of layers), width (number of channels), or image resolution (image size). EfficientNets, on the other hand, perform Compound Scaling - that is, scale all three dimensions while maintaining a balance between all dimensions of the network. We use this trained model for further analysis using Mask-Unmask classification frameworks. To visually inspect what the trained EfficientNet B0 had learned, we implemented GradCam [56] on the network. GradCam assesses which parts of the input image have the highest activation values, given a target class. In this case, we passed the jaw region after facial landmark detection as the input image to the GradCam network for three examples, two masked and one unmasked. The corresponding activation maps are shown in (Figure 3.4).

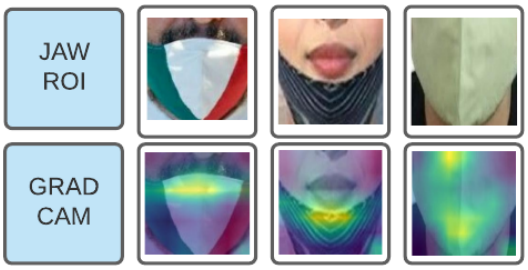

Figure 3.4: GradCam Analysis, showing the activation of different regions on the jaw in the classification model.

Image Size	Encoder	Recall	Accuracy	IOU
100 x 100	Resnet 50	0.96	0.98	0.95
150 x 150	Resnet 50	0.96	0.98	0.95
224 x 224	Resnet 32	0.96	0.98	0.95

Table 3.2: Face Mask Fit Analyser Model Results

3.2 Fit score analysis framework

To determine the fit of the mask on the appropriate region covering the nose and mouth region of the face, we used a semantic segmentation based network (Figure 3.5) and defined a fit score as shown in Equation 3.1. Using the data from Section 2.2, we trained a U-net [53] based model for segmenting images of faces into the mask and unmasked regions. The model uses ResNet 32 and 50 encoders, pre-trained on ImageNet data [26]. The Layers were trained progressively using cyclical learning rates [59]. Different model variations tried using different encoders, and input image sizes are shown in (Table 3.2). Using the output of this model (True Positive(TP) + False Positive(FP)) and the nose-mouth region's facial landmarks, we calculated the fit score of an image of a face. The fit score is the percentage of the area of ROI covered by the mask on the face (Figure 3.5). We employed the Fit score analysis framework on the BLM posts to get an insight into how well people wore masks in groups during the protests.

$$FitScore = \frac{\text{Area Covered by Mask (detected by the segmentation model) of ROI}}{\text{Area of ROI}} * 100 \tag{3.1}$$

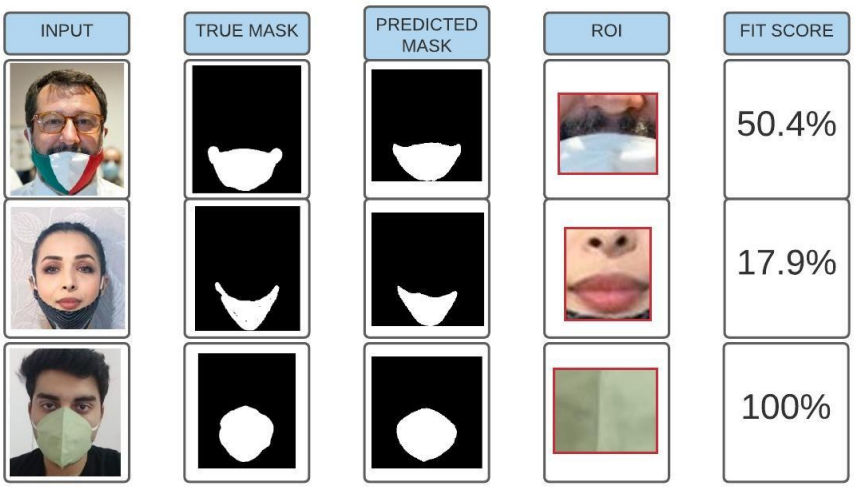

Figure 3.5: Semantic Segmentation Pipeline, showing the True mask, Predicted Mask, Region of Interest (ROI), and Fit Score for the given input image.

Chapter 4

Analysis

4.1 City-wise Analysis

This Section presents the results of experiments conducted to evaluate patterns of people wearing masks in the six cities across the selected time frame, February 1, 2020 and May 31, 2020. A total of 1.66 million faces were detected from all the posts across the six cities. Out of which, a total of 232,706 faces had masks. Table 4.1 shows the City-wise distribution of the detected faces and masks. 1.16 million posts(57% of the total posts collected) had no faces, while 1.89 million (93%) of the total posts had no masked faces. One or more faces were detected in 0.87 million (43%) of the posts, of which 0.61 million(30%) had a single face detected, and 0.26 million (13%) had multiple faces detected. While in 0.14 million(7%) of the total posts, one or more masked faces were detected, out of which 0.12 million(6%) had a single masked face detected.

City	Collected posts	Faces Detected	Masks Detected	% of Faces with Masks
New York City	245,677	200,089	25,413	12.70
Dallas	540,500	444,194	48,119	10.83
Seattle	437,040	312,012	46,019	14.75
Minneapolis	220,999	152,822	30,385	19.88
New Orleans	315,082	321,591	39,420	12.26
Boston	283,757	238,770	43,350	18.15
Total	2,043,055	1,669,478	232,706	13.94

Table 4.1: City-wise distribution of the number of detected Faces, detected Masks and number of Masks per Face through our frameworks.

A group picture is described as a picture with at least two faces detected. Figure 4.1 indicates that as the pandemic grew and the lockdowns came into effect, group posting decreased signif-

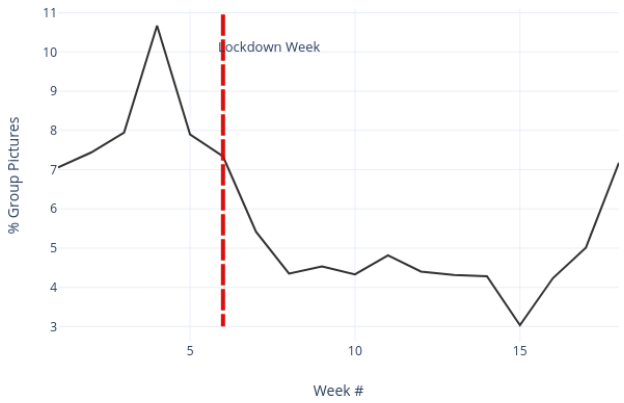

Figure 4.1: Weekly percentages of group pictures detected from all 6 cities between February 1, 2020, and May 31, 2020. A total of 2.04 million posts analysed.

City	Max Lag	Correlation Value	P Values
New York City	1	0.07	.50
Boston	1	0.42	<.001
Minneapolis	6	0.60	<.001
Seattle	1	0.65	<.001
Dallas	1	0.40	<.001
New Orleans	1	0.51	<.001

Table 4.2: Pearson's correlation coefficient and P-values, between Lagged cumulative COVID-19 cases and daily percentages of people wearing masks.

icantly. We also see a sudden increase in the number of group pictures post week 15 (May 16, 2020 - May 22, 2020), which can be linked to the easing lockdown restrictions through the weeks 13,14 and 15 [1, 36, 50, 62].

We studied and compared the number of people wearing masks across the six cities using monthly percentages of masked faces. For each city, the dataset was divided into months, and the percentages of people wearing masks were computed. (Figure 4.2) shows the monthly percentages of mask-wearers for each of the six cities. A general increasing trend is seen across all the cities, suggesting that more people started wearing masks as the pandemic grew in each city.

We also checked for correlation of Covid-19 cases with the percentage of people wearing masks in the cities' respective states. We compared the lagged series of cumulative Covid-19 cases with the percentage of masked photos daily. (Figure 4.3) shows Pearson's and Spearman's correlation for every city with the appropriate lag value for the Covid-19 cases series. Table 4.2 and 4.3 show Pearson's and Spearman's correlation respectively, along with their associated P values.

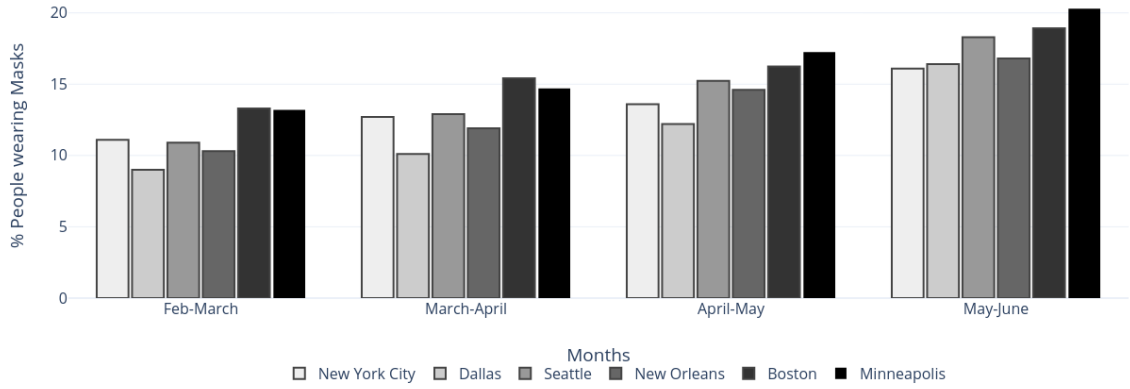

Figure 4.2: Monthly percentages of people wearing masks for each of the six cities, between February 1, 2020 and May 31, 2020. For each city, the dataset was divided into months, and the percentages of people wearing masks were computed.

City	Max Lag	Correlation Value	P Values
New York City	1	0.05	.62
Boston	1	0.60	<.001
Minneapolis	2	0.42	<.001
Seattle	1	0.71	<.001
Dallas	1	0.47	<.001
New Orleans	1	0.54	<.001

Table 4.3: Spearman's correlation coefficient and P values, between Lagged cumulative COVID-19 cases and daily percentages of people wearing masks.

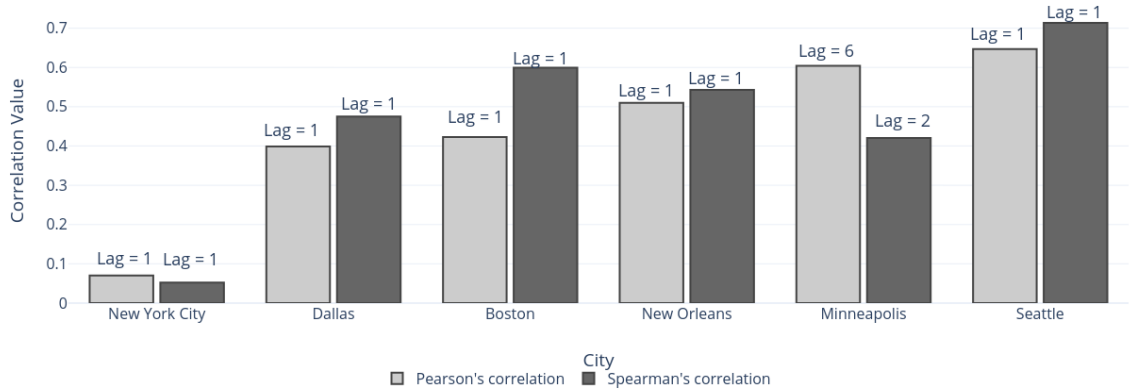

Figure 4.3: Pearson's Correlation and Spearman's Correlation between daily lagged cumulative cases and percentage of masked photos, between February 1, 2020 and May 31, 2020. The length of the series was 120. The lag was selected between 0 and 14 based on the highest correlation value.

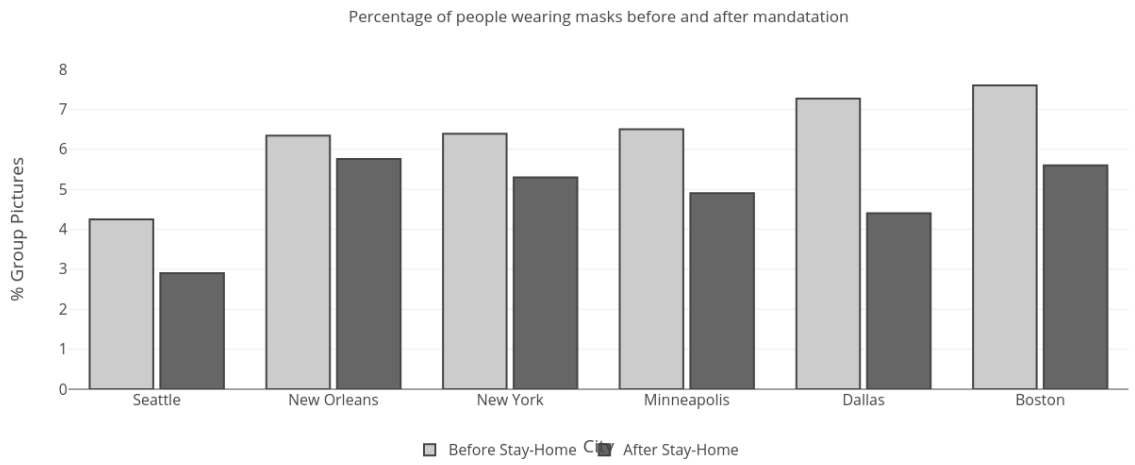

Figure 4.4: Average daily Percentage of Group pictures before and after Stay-at-Home laws for the six cities, between February 1, 2020, and May 31, 2020. A total of 2.04 million posts analysed.

City	Max Lag	Correlation Value	P Values
New York City	1	0.05	.62
Boston	1	0.60	<.001
Minneapolis	2	0.42	<.001
Seattle	1	0.71	<.001
Dallas	1	0.47	<.001
New Orleans	1	0.54	<.001

Table 4.4: Welch's t-test statistic, P values to test for equal means before and after application of stay-at-home laws.

We observe a significant positive correlation, with alpha = 0.01, for all cities except for New York City. This positive relation for both Pearson and Spearman's test indicates that social media behaviour of people changed with the increasing Covid-19 cases. The lag was selected between 0 and 14 based on the highest correlation value. Most of the cities have a maximum correlation at a lag of one day. Therefore, an increase in the number of masks is seen a day after, a spike in the number of cases.

4.2 Impact of Government Policies

This section analyzes how social media behavior changed when wearing masks guidelines and stay-at-home laws were put in place by respective state governments.

City	t statistic	P Values
New York City	1.68	.09
Boston	4.19	<.001
Minneapolis	3.64	<.001

Table 4.5: Welch's t-test statistic, P values to test for equal means before and after application of mask mandates.

4.2.1 Stay At Home Laws

Figure 4.4 shows the average daily percentage of group pictures being posted for each of the six cities before and after the stay-at-home laws were put into place. We see a decrease in the percentages after the stay at home mandates are applied.

1. Boston Department of Public Health issued a two-week stay-at-home advisory on March 23, 2020 following which a drop of 2% was seen in the percentage of group pictures.
2. For Minneapolis, stay-at-home advisory came into effect on March 27, 2020 [8] following which a drop of 1.6% was observed in percentage of group pictures .
3. New Orleans observed a drop of 0.58% in number of group pics, post stay-at-home orders starting March 20, 2020 [10].
4. Dallas saw a drop of 2.8% in the number of group pics after March 23, 2020 [51], after application of stay-at-home orders.
5. Seattle observed a drop of 1.3% after stay-at-home order on May 31, 2020 [5].
6. In New York State state-wide stay-at-home order was declared on March 20, 2020 [8], following which a drop of 1% was seen in percentage group pics.

The average daily percentages of group posting before and after the stay-at-home laws were applied, are statistically different from one another for all the 6 cities with alpha = 0.01. Table 4.4 shows the results of the Welch's t-test for the 6 cities.

4.2.2 Mask Wearing mandates

Figure 4.5 shows the average daily percentage of people wearing masks before and after the state mask mandates were applied for the three cities who implemented these mandates within our selected time range. We see an increase in the percentages after the mandates are put in place for all of the three cities.

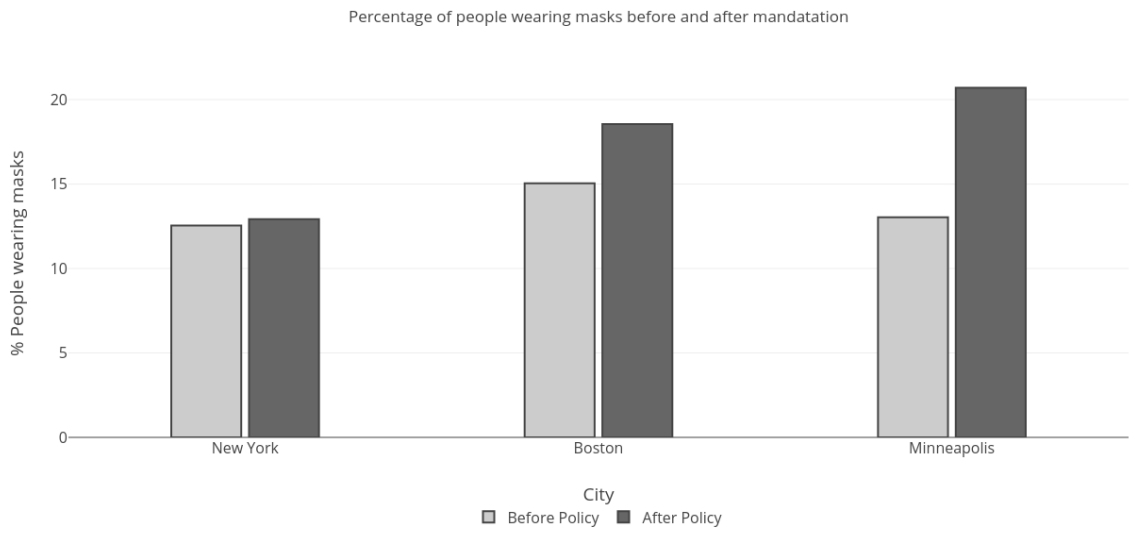

Figure 4.5: Average daily Percentage of people wearing masks, before and after mask-wearing guidelines for New York, Boston, and Minneapolis, between February 1, 2020, and May 31, 2020. There is an increase in the percentage of people wearing masks post the mask-wearing mandates. A total of 750,433 posts analysed.

1. In Boston, residents were asked to wear a mask in public places starting May 6, 2020 [6]. An increase of 3% was seen in the percentage of people wearing masks, in the collected posts for Boston.

2. Mask wearing guidelines were put in place on April 30, 2020 for Minneapolis, encouraging people to wear face coverings [7]. An increase of 7.4% was seen in the percentage of people wearing masks for Minneapolis.

3. In New York State, residents were asked to wear masks in public starting April 15, 2020 [5], following which an increase of around 1% is seen in the percentage of people wearing masks in New York City.

The average daily percentages of people wearing masks before and after the state mask mandates were applied, are statistically different from one another for Boston and Minneapolis with alpha = 0.01, while the difference is not significant for New York City. Table 4.5 shows the results of the Welch's t-test for the 3 cities.

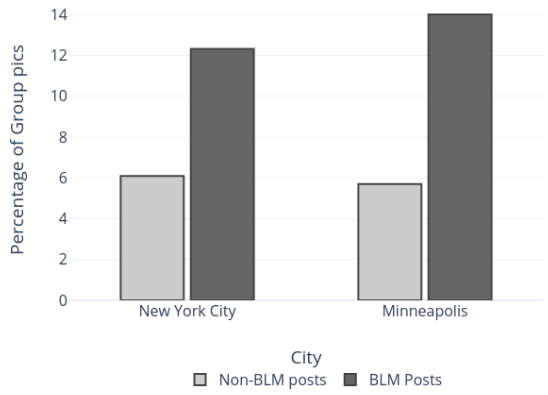

Figure 4.6: Percentage of people wearing masks in groups for BLM and non-BLM posts. More people were wearing masks on average in the protest related posts as compared to the normal posts.

4.3 Protests and the Pandemic

The killing of George Floyd on May 25, 2020 sparked a series of protests [60] and agitation across the country. Protests are designed to stimulate public action for social justice. Such protests involve the physical gathering of people, making it difficult to adhere to social distancing and to wear masks. These protests provide an opportunity to observe how people reacted in terms of wearing masks and social distancing during a pandemic.

Figure 4.6 shows the difference in the percentage of group pics between BLM and non-BLM posts , which implies that a significant number of posts collected under the BLM tags had groups of people present. To get insight into how many people were wearing masks during the protests, we compare non-BLM group pics and BLM-group pics.

Figure 4.7 shows percentages of faces covered with masks in group pictures for Non-BLM, and BLM posts. There was a significant increase in the percentage of people detected with masks, showing that a high percentage of people were abiding to mask-wearing laws during the protests.

We employed the mask-fit analyzer from Section 2 on the BLM posts to get an insight into how well people wore masks in groups during the protests. (Figure 14.8) shows the distribution of the fit scores, for people wearing masks in BLM-posts. Around only 45% of the detected faces had a fit score of $\geq$ 80%. This means that the remaining 55% had some major part of their nose-mouth region not covered.

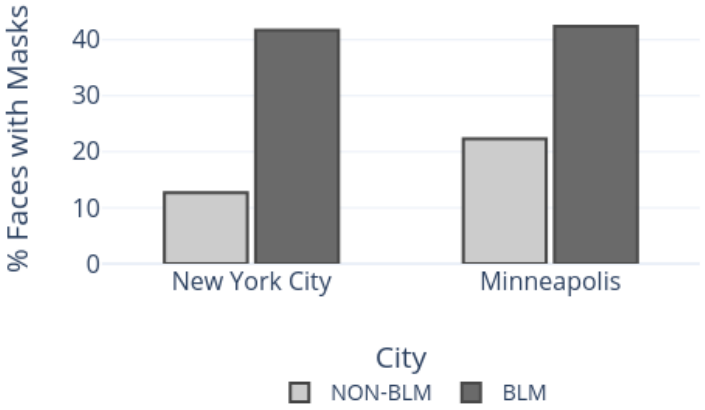

Figure 4.7: Number of Faces v/s Fit Score. Around a quarter of the masked people were wearing the mask with a fit score of greater than equal to 90%

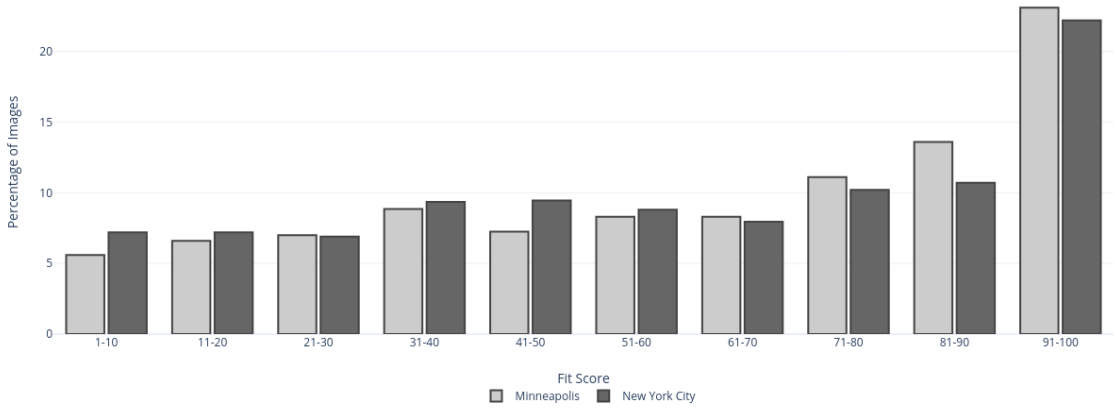

Figure 4.8: Number of Faces v/s Fit Score. Around a quarter of the masked people were wearing the mask with a fit score of greater than equal to 90%

4.4 Democrats vs Republican Analysis

To look at the political conversation going on in the collected data, we plot word clouds of the top political hashtags collected from the captions of posts from the Democratic and the Republican states.

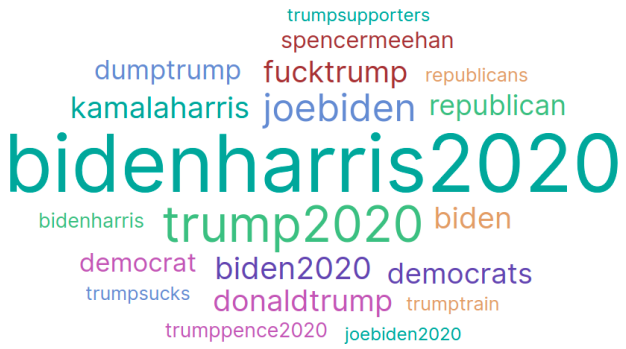

Figure 4.9: Word clouds containing political hashtags, from post captions for Democratic states

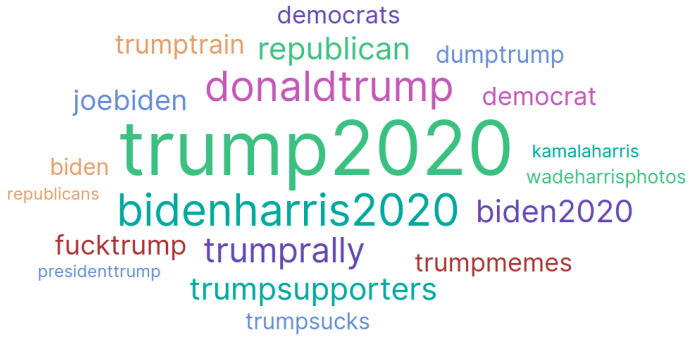

Figure 4.10: Word clouds containing political hashtags, from post captions for Republican states

We observe a higher occurrence of hashtags like 'bidenharris2020', 'trumpsucks', 'dumptrump' in the captions of posts collected from Democratic states, while captions from the Republican states include hashtags like 'trump2020', 'trumpsupporters' in higher numbers. The partisan polarity in these states is captured in the above hashtags effectively.

In order to see how differently the Democratic states and Republican states reacted to the pandemic in terms of face-coverings and social distancing, we plot a timeline of the average percentage of people wearing masks and group pictures. A group picture is defined as a picture with greater than or equal to 2 faces detected. Group pictures are used as an estimator of

adherence to social distancing measures.

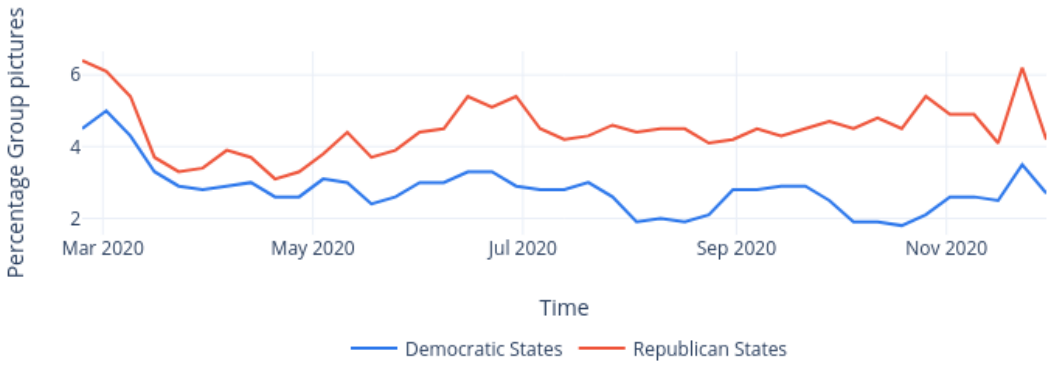

Figure 4.11: Percentage of Group pictures vs time. Republicans states show a higher percentage for the entire time period as compared to Democratic states. The average daily percentage was found to be 2.86% for Democrats while 4.42% for Republicans.

We observe that the percentage of group pictures start at around the same level for both Democratic and Republican states and diverge as the pandemic grows. The Democratic states have a lower percentage of group pictures, indicating a more likely behaviour to follow social distancing measures. The daily percentage value of group pictures were found to be coming from a normal distribution, as seen from the shapiro wilk's test. The difference in the daily average percentage of group pictures for Democrats and Republicans was found to be statistically different from one another with P value much less than 0.001.

Both Democratic and Republican states show a jump in the percentage of masks, at the beginning of the pandemic, but as the pandemic grows Democratic states show a higher percentage of people covering their faces online as compared to Republican states. The daily series of percentage of mask wearers was found to be coming from a normal distribution, due to significant value from the shapiro wilk test. The difference in the daily average percentage of masks detected for Democrats and Republicans was found to be statistically different from one another, with P Value much less than 0.001. This indicates higher mask-wearing among Democrats as compared to Republicans.

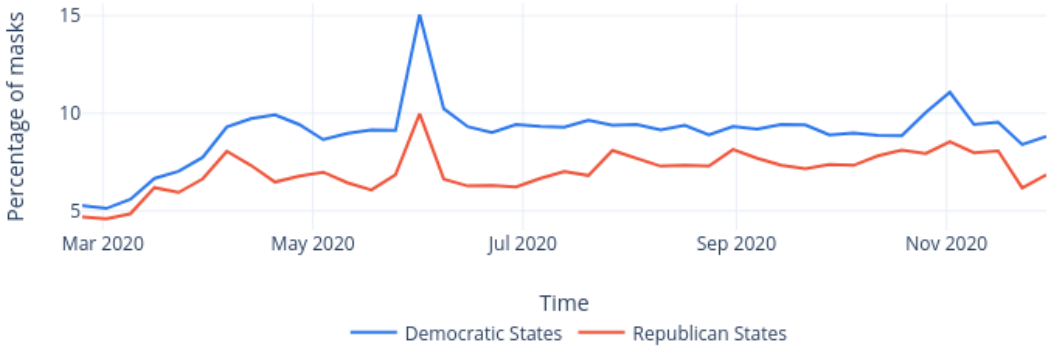

Figure 4.12: Percentage of masks detected vs time. Democratic states show a higher percentage for the entire time period as compared to Republican states. The average daily percentage was found to be 8.41% for Democrats while 6.36% for Republicans.

Chapter 5

Analysis tool

Based on the methodology used till now, we develop a django based web application as an analysis tool for various organisations to see location wise trends in any particular through instagram images. Number of posts, group pictures and number of masked faces can be used as indicators for analysing adherence to various policies.

5.1 Pages

5.1.1 Explore Page

The explore page presents the user with a screen-wide world map with markers placed at different locations. These locations are the ones whose data we have collected and analysed. The user can zoom in and out of the map, and select the marker. The user is redirected to the report page for the particular location, which is explained in Section5.1.3.

5.1.2 Request Page

On the request page the user is presented with a basic form to submit request for the location the user want to see the summary report of. The user is presented with the following fields.

1. Name
2. Email
3. Organisation

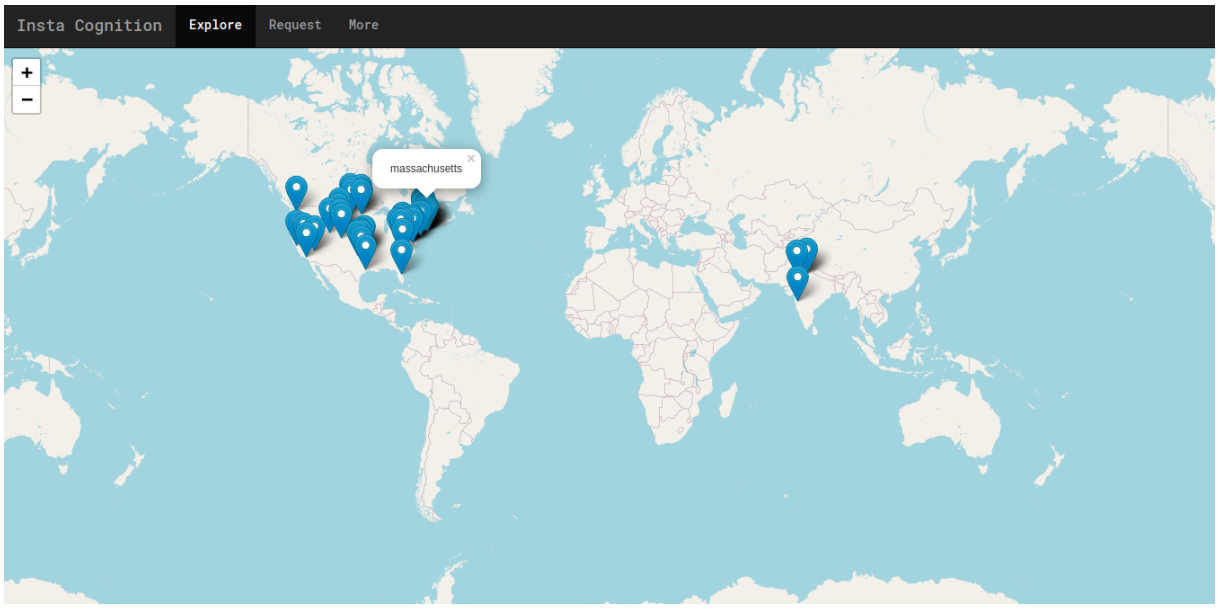

Figure 5.1: The explore page of the dashboard, with world map and location markers.

4. Purpose of Data collection
5. Instagram location page link
6. Start Date of data collection.
7. End Date of data collection.

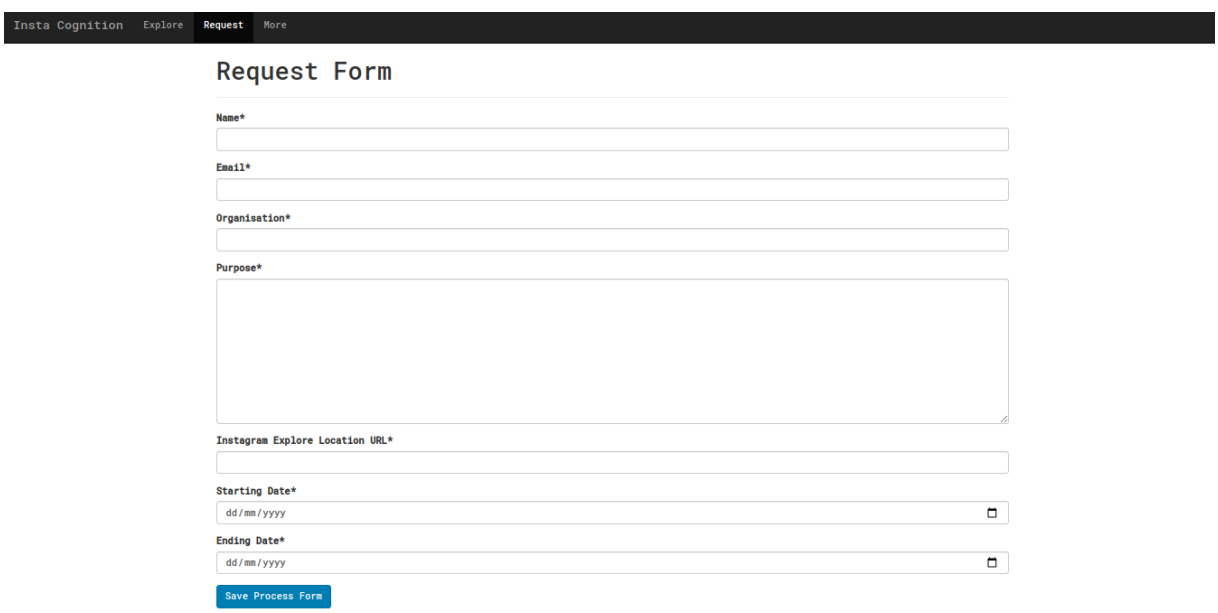

Figure 5.2: The request page of the dashboard.

5.1.3 Report Page

This is the most information dense page in the application. On being redirected from the map marker, A report is opened for the user. The report consists to two sections. The upper section of the report page shows the name of the location and an array of images. The images are a random selection of posts collected from the specific location.

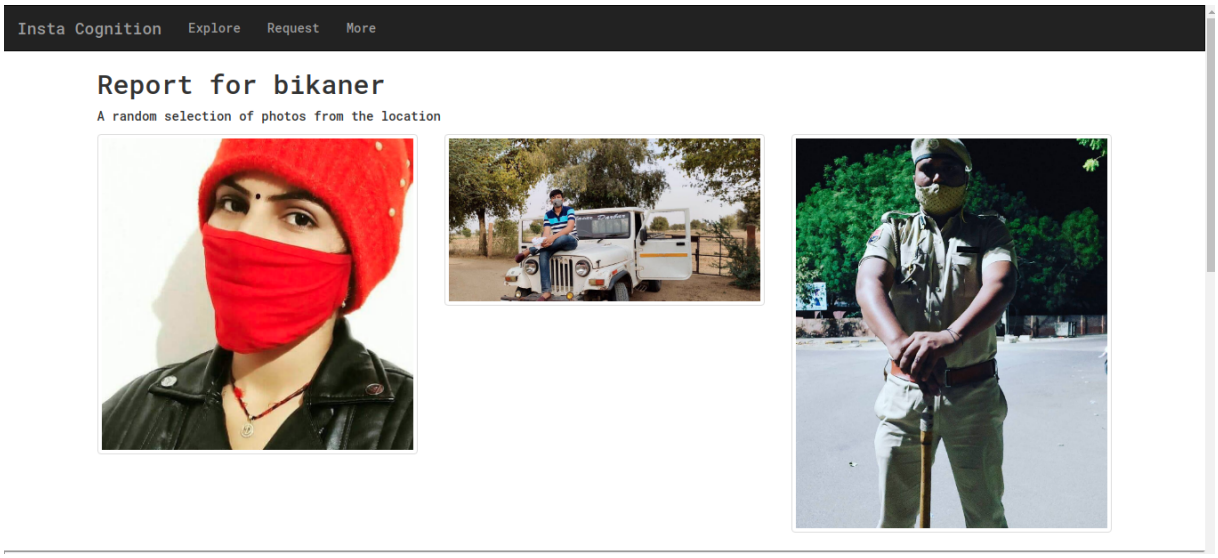

Figure 5.3: Upper section of the report.

The second section of the report contains the snapshot from the Kibana dashboard. The Kibana dashboard consists of the following visualisations.

1. Day wise timeline of the collected posts.
2. Wordcloud of the most occuring hashtags in the post captions.
3. Weekly and daily average percentage of masks.
4. Daily trend of the average number of group pictures.

5.2 Data Pipeline

The following steps are involved in the collection and processing of data in this dashboard.

1. The location is collected from the user.

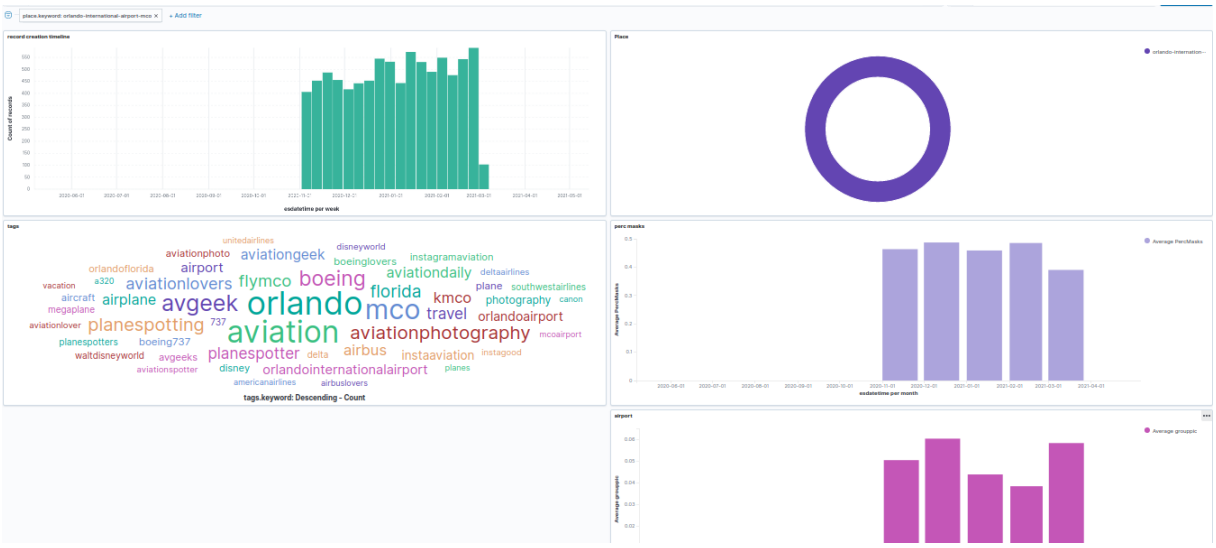

Figure 5.4: Lower section of the report.

2. The Graphql based instagram scraper scrapes the location page within the chose time range.

3. The scraped data and images are stored in a server, the data is stored as jsonls, while the images are sent to the mask classifier.

4. The results from the mask classifier and the original data are combined and saved into the elastic search database.

5. The Kibana dashboard accesses data from the elastic database and the snapshot of the kibana dashboard is sent to the django application.

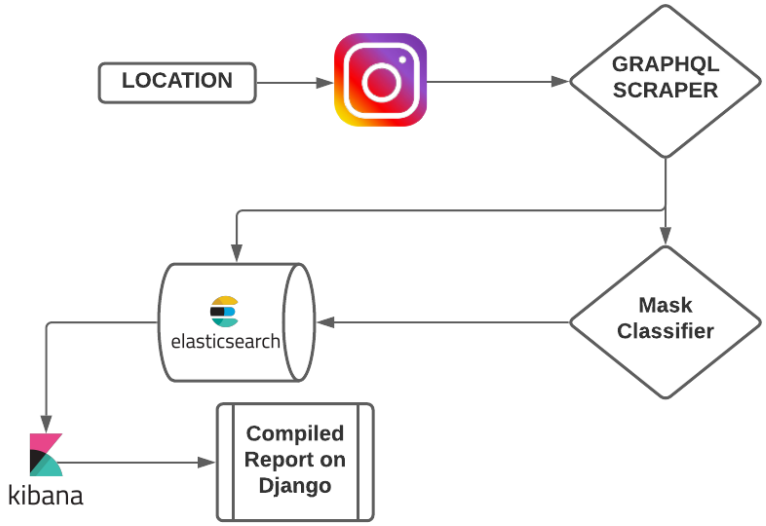

Figure 5.5: Data collection and analysis pipeline for the dashboard.

Chapter 6

Conclusion

The Covid-19 pandemic has given the entire research community and governments a chance to reflect on what kind of system needs to be placed to handle such catastrophes. A renewed focus is emerging in infodemiology [30], especially leveraging mass surveillance data [21,57]. Location tracking [55], periodic self-checks, and image recognition systems have been deployed by many governments [15, 40, 41, 66] to get a handle on the pulse of the pandemic in their states. This study suggests another such handle, which can be applied to certain demographics to achieve similar near-real-time tracking of the pandemic's spread. Instagram and other social media platforms have been very successful in tracking the number of visitations to public places [61]. In the context of Covid-19, public places are the focal points for the spread of viruses. It has been well-documented that face masks and social distancing are the two most effective non-pharmaceutical interventions (NPIs) to curb the spread of Covid-19 [52]. However, the use of masks and effective social distancing are often self-reported [43] , without any proof to corroborate the claims. Models built on image data with location data [44] can be a powerful tool for the authorities to keep track of the pandemic's pulse.

Tools built on images, as show in 5 with location data can be a powerful tool for the authorities to keep track of the pandemic's pulse. This study examines 2.04 million posts collected from six cities from USA between February 1, 2020, and May 31, 2020, for adherence of mask-wearing and social distancing as seen through social media. This study finds a general increasing trend in mask-wearing and a decreasing trend of group pictures as the pandemic grew. The stay-at-home laws showed a significant drop in group posting for all six cities, while the mask mandates showed a significant increase in mask wearing for two of the three cities analysed. Although these results suggest an upward trend in adoption of preventive methods, a large portion of non-

adopters seen online, indicate a need for supplemental measures to increase the effectiveness of such methods. Posts related to protests were found to capture the lack of attention given to safety measures through high percentage of detected group pictures, and incorrect mask-wearing. The methodology used provides a directional indication of how government policies can be indirectly monitored. These findings can help governments and other organisations as indicators for a successful implementation of non-pharmaceutical interventions for the pandemic. Our analysis of Instagram finds significant differences between Democratic and Republican states in reaction to non-pharmaceutical interventions against COVID-19. Our findings support the existing on-ground and self-reported surveys, showing significantly higher adoption of face-coverings and social distancing, by Democratic states as compared to Republican states. This goes on to show the relevance of Social Media as a critical tool for understanding and tracking real-world events and sentiments of the population.

6.1 Limitation

The images present in datasets, VAMA-C and VAMA-S, are high definition and in RGB mode. If the trained models have to be deployed in a new setting like CCTV feed, certain image augmentation techniques like grayscaling and rescaling might be needed for fine-tuning the model. For the analysis, we choose the image data from six major US cities by population and correlate it with state-wide Covid-19 cases. Since these cities are some of the most populated cities of the respective states, it is reasonable to assume that they will be the hot-beds of the Covid-19 spread in that state. The analysis has been conducted images fetched from the social-media platform Instagram. The authors understand that these images might not be a representation of the entire city population [16]. However, they do represent a wide demographic of internet users [17]. With recent studies conducted on geo-tagged text data present in Instagram posts [54], our assumption of a fair population representation in the Instagram data might not be too far-fetched. Among the mask wearers the celebrity posts(posts with likes greater than 10000) contributed to only 0.2% of the total posts which contained at least one mask, which shows that major contribution to the above collected data is through the general population. However, the authors do acknowledge that capturing meta-data for users while performing similar studies might yield a conclusive answer to the question of fair representation. Capturing and using meta-data can be a future work, building on these results.

6.2 Future Works

An addition to the modeling pipeline would be an indoor/ outdoor environment detector, similar to Zhou, et al. [65]. Another addition to the analysis could be selectively looking at specific user activity who are deemed as "influencers" on the network. Their activity on the network can be analysed in conjunction with the activity of their followers. This can help determine the role of social networks and the power of certain influential nodes in that network over other people's behavior during critical times such as a pandemic. A natural extension of this work is its replication across different social media platforms like Twitter, Facebook, Baidu.

6.3 FAIR

The gathered data consists of publicly available information about a social network. Gathering and examining of this data provided significant insights into public health recommendations. Our dataset also conforms to the Fairness, Accessibility, Interoperability, and Reusability of the collected dataset (FAIR) principles. In particular, the datasets used for training the models, are "findable", as it is shared publicly at [3]. This dataset is also "accessible", given the format used (.png) is popular for data transfer and storage. This file format also makes the data "interoperable" and "reusable", given that most programming languages and softwares have libraries to process image files. The data was collected through public API endpoints of Instagram. The data we collected was stored in a central server with restricted access and firewall protection.

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